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HACKING MOBILE PLATFORM

MODULE 17

VYANKATESH ANDIL



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1: MOBILE PLATFORM HACKING

1: Mobile Platform Hacking

1. INTRODUCTION TO MOBILE PLATFORM HACKING

Mobile platform hacking focuses on **identifying, exploiting, and mitigating vulnerabilities** in mobile devices, operating systems, applications, and communication channels.

With the rapid adoption of smartphones, mobile devices have become **prime targets** for attackers due to:

- Sensitive personal data
 - Financial applications
 - Always-on connectivity
 - BYOD environments
-

2. OBJECTIVES OF MOBILE PLATFORM HACKING

The primary objectives include:

- Gaining unauthorized access to mobile devices
- Extracting sensitive data (contacts, SMS, credentials)
- Bypassing authentication mechanisms
- Installing malicious applications
- Monitoring user activity

From a **CEH perspective**, the goal is to **understand attack techniques to better defend mobile environments**.

3. MOBILE THREAT LANDSCAPE

3.1 Why Mobile Devices Are High-Value Targets

- Store personal & corporate data
 - Used for authentication (OTP, biometrics)
-

1: MOBILE PLATFORM HACKING

- Often poorly patched
 - Users install third-party apps
-

3.2 COMMON MOBILE ATTACK VECTORS

- Malicious applications
 - Insecure Wi-Fi connections
 - SMS & MMS attacks
 - Bluetooth exploitation
 - Phishing & social engineering
-

4. MOBILE OPERATING SYSTEMS OVERVIEW

4.1 Android Platform

- Open-source (Linux-based)
- Highly customizable
- Larger attack surface
- Dominant market share

4.2 iOS Platform

- Closed-source ecosystem
 - Strict app sandboxing
 - Strong security controls
 - Limited attack surface compared to Android
-

5. MOBILE DEVICE ARCHITECTURE

5.1 Core Components

- Kernel

1: MOBILE PLATFORM HACKING

- Hardware abstraction layer
 - Runtime environment
 - Application framework
-

5.2 Security Mechanisms

- Sandboxing
 - Permission model
 - Secure boot
 - Application signing
-

6. MOBILE APPLICATION ECOSYSTEM

6.1 Application Sources

- Official app stores
 - Third-party app stores
 - Sideloaded applications
-

6.2 Risks

- Malware injection
- Trojans disguised as legitimate apps
- Repackaged applications

1: MOBILE PLATFORM HACKING

7. MOBILE PLATFORM ATTACK SURFACE

Layer	Attack Surface
Hardware	Sensors, baseband
OS	Kernel vulnerabilities
Apps	Insecure coding
Network	Wi-Fi, Bluetooth, NFC
User	Phishing, social engineering

8. RISKS & IMPACT OF MOBILE PLATFORM ATTACKS

- Identity theft
- Financial fraud
- Corporate data leakage
- Device takeover
- Surveillance & spying

9. MITRE ATT&CK MAPPING

Area	MITRE Tactic	Technique ID	Description
Mobile Discovery	Discovery	T1420	File and Directory Discovery
Credential Theft	Credential Access	T1414	Credential Access
Malicious App	Initial Access	T1475	Deliver Malicious App

1: MOBILE PLATFORM HACKING

10. CEH EXAM-ORIENTED NOTES

- Mobile hacking ≠ only apps
- Android = larger attack surface
- iOS = restrictive but not immune
- App permissions are critical
- BYOD increases risk

2: MOBILE THREATS, MALWARE TYPES & ATTACK VECTORS

2: Mobile Threats, Malware Types & Attack Vectors

1. INTRODUCTION TO MOBILE THREATS

Mobile threats exploit **device vulnerabilities, user behavior, insecure applications, and wireless communication channels**. Attackers aim to gain **control, steal data, or spy on users**.

2. MOBILE MALWARE

2.1 Definition

Mobile malware is malicious software designed to target **smartphones and tablets**.

2.2 Types of Mobile Malware

a) Trojans

- Disguised as legitimate apps
- Steal credentials or install backdoors

b) Spyware

- Monitors calls, messages, GPS
- Operates silently

c) Ransomware

- Locks device or encrypts data
- Demands ransom

d) Adware

- Displays unwanted ads
- May collect user data

e) Mobile Botnets

- Infected devices controlled remotely
- Used for DDoS or spam

2: MOBILE THREATS, MALWARE TYPES & ATTACK VECTORS

3. MOBILE ATTACK VECTORS

3.1 Malicious Applications

- Fake apps
 - Repackaged apps
 - Trojanized games
-

3.2 SMS & MMS Attacks

- Malicious links
 - Premium SMS fraud
-

3.3 Phishing Attacks

- Fake banking apps
 - Credential harvesting
-

3.4 Network-Based Attacks

- Rogue Wi-Fi hotspots
 - Man-in-the-Middle attacks
-

4. RISKS AND IMPACT

- Financial theft
- Identity compromise
- Corporate espionage
- Device surveillance

2: MOBILE THREATS, MALWARE TYPES & ATTACK VECTORS

5. MITRE ATT&CK MAPPING

Attack	Tactic	Technique ID
Malicious App	Initial Access	T1475
SMS Phishing	Initial Access	T1438
Spyware	Collection	T1417

6. CEH Exam Notes

- Most mobile malware = Trojan-based
- Third-party app stores increase risk
- SMS phishing = smishing

3: ANDROID PLATFORM HACKING

3: Android Platform Hacking

1. ANDROID ARCHITECTURE

1.1 Key Layers

- Linux Kernel
 - Android Runtime
 - Application Framework
 - Applications
-

2. ANDROID SECURITY MODEL

- Application sandboxing
 - Permission-based access
 - Application signing
-

3. ANDROID ATTACK TECHNIQUES

3.1 Malicious APKs

- Trojanized applications
 - Reverse-engineered and modified apps
-

3.2 Rooting Attacks

- Exploit kernel vulnerabilities
 - Gain superuser access
-

3.3 Insecure Permissions

- Excessive permissions
 - Permission misuse
-

3: ANDROID PLATFORM HACKING

3.4 Insecure Storage

- Sensitive data stored in plaintext
-

4. ANDROID EXPLOITATION FLOW

1. App analysis
 2. Reverse engineering
 3. Payload injection
 4. Deployment
-

5. RISKS

- Complete device compromise
 - Persistent malware
-

6. MITRE ATT&CK MAPPING

Activity	Tactic	Technique
Rooting	Privilege Escalation	T1404
Malicious APK	Initial Access	T1475
Data Theft	Collection	T1417

7. CEH Exam Notes

- Android = open ecosystem
- Rooting bypasses sandbox
- Permissions are critical

4: IOS PLATFORM HACKING

4: iOS Platform Hacking

1. IOS SECURITY ARCHITECTURE

- Secure boot chain
 - Code signing
 - App sandboxing
-

2. JAILBREAKING IOS

2.1 Definition

Jailbreaking removes Apple's restrictions to:

- Install unauthorized apps
 - Access system files
-

2.2 Risks

- Security controls disabled
 - Malware risk increases
-

3. IOS ATTACK TECHNIQUES

3.1 Malicious Profiles

- Configuration profile abuse

3.2 Spyware Installation

- Surveillance tools

3.3 iCloud Attacks

- Credential compromise

4: IOS PLATFORM HACKING

4. RISKS

- Privacy violation
 - Data exfiltration
-

5. MITRE ATT&CK MAPPING

Activity	Tactic	Technique
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Jailbreak	Privilege Escalation	T1404
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Spyware	Collection	T1417
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6. CEH Exam Notes

- iOS is more restrictive
- Jailbreaking weakens security
- Code signing is key defense

5: MOBILE PLATFORM HACKING TOOLS

5: Mobile Platform Hacking Tools

1. INTRODUCTION TO MOBILE HACKING TOOLS

Mobile hacking tools help security professionals and attackers to:

- Analyze mobile applications
- Exploit OS-level vulnerabilities
- Intercept mobile communications
- Extract sensitive data

CEH focuses on **tool purpose, methodology, and security impact**, not only usage.

2. ANDROID DEBUG BRIDGE (ADB)

2.1 Overview

ADB is a command-line tool used to **communicate with Android devices**.

2.2 Objectives

- Install/uninstall applications
 - Access device shell
 - Extract logs and files
-

2.3 Attack Usage

- Abuse enabled USB debugging
 - Extract sensitive files
 - Deploy malicious apps
-

2.4 Risks

- Full device access if unlocked

5: MOBILE PLATFORM HACKING TOOLS

- Data extraction
-

2.5 Mitigation

- Disable USB debugging
 - Lock devices
 - Enforce device encryption
-

2.6 MITRE Mapping

- **T1404 – Privilege Escalation**
 - **T1417 – Data Collection**
-

2.7 CEH Exam Notes

- ADB requires physical or debug access
 - Common in Android forensic and hacking labs
-

3. DROZER

3.1 Overview

Drozer is an **Android security assessment framework**.

3.2 Objectives

- Identify insecure app components
 - Exploit IPC vulnerabilities
-

3.3 Methodology

1. App enumeration
2. Component analysis

5: MOBILE PLATFORM HACKING TOOLS

3. Permission abuse

3.4 Risks

- Data leakage
 - Unauthorized component access
-

3.5 Mitigation

- Secure IPC mechanisms
 - Validate permissions
-

3.6 MITRE Mapping

- **T1475 – Malicious App**
 - **T1417 – Collection**
-

3.7 CEH Exam Notes

- Drozer targets **Android apps**, not OS
-

4. METASPLOIT (MOBILE EXPLOITATION)

4.1 Overview

Metasploit provides payloads and exploits for **Android and iOS platforms**.

4.2 Objectives

- Gain shell access
- Deploy spyware
- Validate vulnerabilities

5: MOBILE PLATFORM HACKING TOOLS

4.3 Risks

- Device instability
 - Detection
-

4.4 Mitigation

- OS patching
 - Restrict app installation
-

4.5 MITRE Mapping

- **T1475 – Initial Access**
 - **T1414 – Credential Access**
-

5. MOBILE SPYWARE TOOLS

Examples:

- SpyNote
- AndroRAT

Used for:

- Keylogging
 - SMS interception
 - GPS tracking
-

Risks

- Severe privacy invasion
- Legal consequences

5: MOBILE PLATFORM HACKING TOOLS

6. TOOL SUMMARY TABLE

Tool	Platform	Purpose
ADB	Android	Device access
Drozer	Android	App testing
Metasploit	Android/iOS	Exploitation
Spyware	Both	Surveillance

6: MOBILE PLATFORM SECURITY, COUNTERMEASURES & HARDENING

6: Mobile Platform Security, Countermeasures & Hardening

1. INTRODUCTION TO MOBILE HARDENING

Mobile hardening reduces the **attack surface** by securing:

- OS
 - Applications
 - Network connectivity
-

2. MOBILE OS HARDENING

2.1 Android Hardening

- Disable USB debugging
 - Install apps only from trusted stores
 - Regular OS updates
-

2.2 iOS Hardening

- Avoid jailbreaking
 - Enable auto-updates
 - Use strong device passcode
-

3. APPLICATION-LEVEL SECURITY

- Least-privilege permissions
- Secure data storage
- Code obfuscation

6: MOBILE PLATFORM SECURITY, COUNTERMEASURES & HARDENING

4. NETWORK SECURITY

- Avoid public Wi-Fi
 - Use VPNs
 - Disable unused wireless interfaces
-

5. MOBILE DEVICE MANAGEMENT (MDM)

5.1 Role

- Enforce policies
 - Remote wipe
 - App control
-

6. USER AWARENESS

- Avoid unknown links
 - Verify app permissions
 - Recognize phishing
-

7. INCIDENT RESPONSE FOR MOBILE ATTACKS

1. Detect suspicious activity
2. Isolate device
3. Remove malware
4. Reset credentials

6: MOBILE PLATFORM SECURITY, COUNTERMEASURES & HARDENING

8. MITRE ATT&CK MAPPING

Defense Area	Tactic	Technique
Hardening	Defense Evasion	T1627
MDM	Mitigation	Defensive Controls
Monitoring	Detection	Defensive Controls

9. CEH Exam Notes (Part 6)

- Mobile hardening = OS + user
- MDM is critical in enterprises
- Jailbreaking increases risk
- VPN protects network traffic

REAL-WORLD ATTACK SCENARIOS, CASE STUDIES & SUMMARY

Real-World Attack Scenarios, Case Studies & Summary

1. REAL-WORLD MOBILE ATTACK SCENARIOS

1.1 Scenario 1: Malicious Android Banking App

Attack Flow

1. Attacker creates a fake banking APK
2. Distributes it via third-party app store
3. User installs and grants excessive permissions
4. Credentials are exfiltrated

Impact

- Financial fraud
- Identity theft

Mitigation

- Install apps only from trusted sources
 - Verify app permissions
-

1.2 Scenario 2: Spyware via Phishing SMS (Smishing)

Attack Flow

1. User receives malicious SMS link
2. Link installs spyware APK
3. Attacker monitors calls, SMS, GPS

Impact

- Privacy breach
- Surveillance

Mitigation

- User awareness

REAL-WORLD ATTACK SCENARIOS, CASE STUDIES & SUMMARY

- Mobile security solutions
-

1.3 Scenario 3: iOS Jailbreak Exploitation

Attack Flow

1. Device is jailbroken
2. Security controls disabled
3. Malicious configuration profile installed

Impact

- Data leakage
- Persistent compromise

Mitigation

- Avoid jailbreaking
 - MDM enforcement
-

2. MOBILE INCIDENT RESPONSE LIFECYCLE

2.1 Detection

- Battery drain
- Abnormal network traffic
- Unexpected app behavior

2.2 Containment

- Isolate device
- Disable network access

2.3 Eradication

- Remove malicious apps
- OS reinstallation

REAL-WORLD ATTACK SCENARIOS, CASE STUDIES & SUMMARY

2.4 Recovery

- Restore from trusted backup
 - Change credentials
-

3. MOBILE ATTACK KILL CHAIN (CEH PERSPECTIVE)

Phase	Activity
Reconnaissance	OS identification
Weaponization	Malicious APK
Delivery	App store / SMS
Exploitation	Permission abuse
Installation	Spyware
C2	Remote control
Actions	Data exfiltration

4. COMMON MOBILE MISCONFIGURATIONS

- USB debugging enabled
- Excessive app permissions
- Outdated OS
- Jailbroken/rooted devices
- No screen lock

REAL-WORLD ATTACK SCENARIOS, CASE STUDIES & SUMMARY

5. COMPLETE MITRE ATT&CK MAPPING

Phase	Tactic	Technique ID	Description
Initial Access	Initial Access	T1475	Deliver Malicious App
Execution	Execution	T1406	Native API
Persistence	Persistence	T1627	Install Malicious Profile
Collection	Collection	T1417	Capture Data
Exfiltration	Exfiltration	T1418	Exfiltration Over Network

6. CEH Exam Strategy – Module 17

6.1 High-Probability Exam Topics

- Android vs iOS security
- Rooting vs Jailbreaking
- Mobile malware types
- ADB & Drozer
- MDM role

6.2 Trick Concepts

- Jailbreaking ≠ malware (but increases risk)
- Mobile hacking ≠ only apps
- Permissions misuse is a major issue

7. Quick Revision Table

REAL-WORLD ATTACK SCENARIOS, CASE STUDIES & SUMMARY

Concept	Key Point
Smishing	SMS-based phishing
Rooting	Android privilege escalation
Jailbreaking	iOS security bypass
Drozer	Android app assessment
MDM	Enterprise mobile security

8. Best Practices for Mobile Security

- Keep OS updated
- Use official app stores
- Strong device lock
- Avoid public Wi-Fi
- Enable remote wipe

CONCLUSION

Mobile platform hacking highlights how **devices, applications, networks, and users** combine to form a complex attack surface. Attackers exploit **malicious apps, misconfigurations, and user trust**, while defenders must rely on **hardening, monitoring, and awareness**.

Mastering this module prepares you for:

- CEH examination
 - Mobile penetration testing
 - Enterprise mobile security
-

REAL-WORLD ATTACK SCENARIOS, CASE STUDIES & SUMMARY

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